Multimodal Nowcasting of Sparse Charge-Exchange Spectroscopy on KSTAR: Fast Diagnostics Recover Ion Temperature Beyond Future-Using Interpolation, but Not Toroidal Rotation

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Abstract

Charge-exchange spectroscopy (CES) is the primary measurement of ion temperature (T_i) and toroidal rotation (V_{rot}) in tokamaks, but photon-collection constraints make it slow and intermittently missing: on the KSTAR 10 ms grid studied here, 8.2% of T_i and 23.9% of V_{rot} samples are absent, the two targets go missing independently, and a further 41.1% of V_{rot} rows are instrument holds, leaving 65.0% of the grid without independent V_{rot} information. We ask whether the always-available fast diagnostics—beam-emission spectroscopy (BES), electron-cyclotron-emission imaging (ECEI), and Mirnov coils—together with the irregular past CES history can recover information at missing CES timesteps that *temporal interpolation of CES alone cannot.* Because the fast diagnostics are dense on the 10 ms grid, we frame the problem as *causal full-grid sequence estimation*: a compact (0.36M-parameter) two-branch recurrent nowcaster consumes every grid row as context, handles the sparse targets in a per-target masked loss rather than by dropping rows, and routes the V_{rot} branch away from the fast diagnostics by construction; a windowed GRU–attention model with observation-masked pooling is the paired control. Evaluation is deliberately adverse: offline interpolation (linear, monotone-cubic PCHIP, local AR, and a Gaussian process) may see *past and future* CES neighbours, while the model is strictly causal, and a *causal* GP represents the strongest deployable baseline. Under a pre-registered protocol—plateau-minimal history (W=2), hold-free training and scoring, file-level splits, a test set never touched by model selection, a shot-clustered paired bootstrap (10,000 resamples), and *two co-primary populations* that treat the 0.53% of T_i spectral-fit failures (>3 keV) as missing or keep them, with an unqualified claim required to hold in both—the nowcaster beats future-using PCHIP on T_i with skill +0.17 to +0.26 (spike-cut) and +0.23 to +0.32 (spike-inclusive), 95% CIs excluding zero on *four independent held-out splits in both populations*, and beats the causal GP on all eight; V_{rot} clears the same gate on 1/4 and 2/4 splits and is reported as a tie. Two stress tests separate the offline from the deployment claim: reweighting the scored points onto the covariate distribution of the *genuinely missing* ones keeps the advantage over every causal method on all splits and over PCHIP on 2/4 (cut) and 4/4 (inclusive); re-splitting strictly in time across campaigns—which removes the windowed control's advantage over interpolation—leaves the sequence nowcaster ahead of PCHIP and of the causal GP on 4/4 initialisations in both populations. The edge concentrates in high-variability neighbourhoods (peak-stratum T_i skill +0.45 to +0.72, 8/8 PASS); calibrated conformal intervals beat both baselines' in every cell; a 21k-parameter latent-bottleneck model with an exact persistence-plus-correction decomposition matches the backbone's T_i skill under the spike cut, and widening the backbone from 34k to 879k parameters changes it by less than ±0.01—the information, not the estimator, is the ceiling. Input ablations show the $T_i \leftrightarrow V_{\mathrm{rot}}$ asymmetry is physical: history alone falls below interpolation for T_i in the cut population, while zeroing the fast diagnostics leaves V_{rot} bit-identical (unobserved NBI torque; 100 Hz-sampled Mirnov signals alias away kHz mode rotation). We release the full pipeline, pre-registration, and evaluation artifacts.

1 Introduction

Ion temperature T_i and toroidal rotation V_{rot} at the plasma edge govern key elements of tokamak performance and stability, including pedestal structure, ELM behavior, and access to high-confinement regimes. The workhorse measurement of both quantities is charge-exchange spectroscopy (CES) [1], which infers T_i from Doppler broadening and V_{rot} from the Doppler shift of charge-exchange line emission. Because CES must integrate photons to reach adequate signal-to-noise, it is intrinsically slow, and individual time points are frequently lost to exposure and signal-quality problems. In the KSTAR data studied here, even on a uniform 10 ms grid, 8.2% of T_i and 23.9% of V_{rot} observations are missing—*independently* of each other—while a further 41.1% of V_{rot} rows repeat the previous reading bit-for-bit (§3.4), so only 35.0% of the grid carries an independent V_{rot} measurement. Empirically in our dataset, the missingness concentrates at physically interesting moments (low signal-to-noise phases, ELMs, transitions), i.e. it is missing-not-at-random in the sense of Little and Rubin [2] (MNAR).

Meanwhile, several fast diagnostics observe the same plasma without gaps: BES measures density-fluctuation structure, ECEI images electron temperature, and Mirnov coils record magnetic fluctuations. None of them measures T_i or V_{rot} directly. The question this paper asks is:

> *At a 10 ms timestep where CES is missing, can the simultaneous fast diagnostics plus the irregular past CES history recover T_i and V_{rot} information that temporal interpolation of CES alone cannot?*

A positive answer would justify a data-driven *virtual sensor*: a model that fills CES gaps online, without the strong inverse-mapping assumptions required by hardware alternatives, enabling gap-free pedestal analysis and, ultimately, real-time use.

A deliberately hard bar. The obvious baseline for gap-filling is interpolation of the CES channel itself. We make the comparison intentionally adverse to the model: the interpolation baselines (linear, monotone-cubic PCHIP [3], a local AR reference, and a Gaussian process) are *offline*—they may use CES neighbours on *both sides* of the target time, including the future—whereas the model is strictly causal, seeing only the fast diagnostics up to the target step and the past CES history. If a causal model beats a future-using interpolant, the fast diagnostics must carry CES-relevant information that the temporal structure of CES alone does not contain; and a model that beats future-using interpolation a fortiori beats every causal baseline. Because the strongest *deployable* competitor is not persistence but a past-only smoother, we also add a *causal* GP arm and report the model against it separately.

Contributions.

1. **A causal full-grid framing for sparse-target nowcasting** (§4). The fast diagnostics are dense on the grid, so instead of assembling per-sample windows and dropping unlabelled rows we keep every input-complete row as sequence context and move the sparsity into the loss: a two-branch causal LSTM reads the whole past of a discharge segment with per-target carried-value, staleness and observation-flag channels, and the V_{rot} branch is *structurally* blocked from the fast diagnostics. Against a paired windowed GRU–attention control with observation-masked pooling, this reframing is worth $+0.081$ pooled T_i skill (16 runs, run-cluster CI $[+0.067, +0.096]$, 16/16 positive) at a fraction of the training cost—and it is the reach it buys that clears the causal GP.

2. **A pre-registered, two-population, shot-clustered evaluation protocol** (§5). History length is set by a plateau-minimal rule from a 24-run sweep ($W{=}2$); instrument holds are removed from training and scoring; the 0.53% of T_i spectral-fit failures (>3 keV) define two co-primary populations—treated as missing in every arm, or kept everywhere—and *an unqualified claim must hold in both.* The test set is never read by model selection, significance is a shot-clustered paired bootstrap, and every model decision was fixed in writing before the corresponding TEST scoring.

3. **A robust positive result for T_i** (§6): the causal nowcaster beats future-using PCHIP with $\mathrm{skill}_{\mathrm{PCHIP}} = +0.17$ to $+0.26$ (cut) and $+0.23$ to $+0.32$ (inclusive), CIs excluding zero on all four held-out splits in both populations, and beats the causal GP on all eight split–population cells; it ties the offline GP, the strongest future-using smoother.

4. **A physics-grounded negative result for V_{rot}**, reported as a finding rather than a failure: V_{rot} ties future-using interpolation (1/4 and 2/4 significant, what noise alone would give) while beating persistence on most splits; zeroing the fast diagnostics leaves the V_{rot} output bit-identical and zeroing history collapses it, so rotation remains history-autocorrelation-limited at 10 ms (unobserved NBI torque; Mirnov aliasing). We call this the $T_i \leftrightarrow V_{\text{rot}}$ *information asymmetry*.

5. **Two stress tests that separate two claims usually conflated** (§6.5, §6.6). Reweighting the scored points onto the covariate distribution of the genuinely missing ones keeps the T_i advantage over every causal method on 4/4 splits in both populations and over PCHIP on 2/4 (cut) and 4/4 (inclusive). Re-splitting strictly in time across campaigns removes the windowed control's advantage over interpolation (2/4 and 0/4)—the fast diagnostics drift $1.22\,\sigma$ (BES) between periods against $0.115\,\sigma$ for the targets—while the sequence nowcaster, which standardizes each discharge's inputs by its own statistics, stays ahead of PCHIP and of the causal GP on 4/4 initialisations in both populations.

6. **Where the edge lives**: in high-variability neighbourhoods (peak-stratum T_i skill +0.45 to +0.72, PASS on 8/8 population–split cells; V_{rot} +0.54 to +0.79 point estimates, nil in the bulk) and into the non-adjacent regime ($\Delta t > 15$ ms pooled, PASS in both populations).

7. **The complexity ladder and the size axis, both measured** (§6.9): a 21,498-parameter model that predicts *persistence plus a bounded correction read linearly from eight probeable latents* matches the backbone's T_i skill under the spike cut (paired mean +0.002)—and loses −0.19 without it, because a bounded correction cannot recover a spiked anchor—while widening the backbone from 34k to 879k parameters moves T_i skill by less than ± 0.01. The input set, not the estimator, is the ceiling; the levers that remain are named (§9).

8. **Deployability, measured** (§8): distribution-free conformal intervals that beat both baselines' intervals in all 32 population–target–variant–split cells without touching the predictor, and stateful one-step inference of the recurrent nowcaster inside the 10 ms cadence on a CPU.

2 Related work

Profile fitting and gap-filling in fusion diagnostics. Gaussian-process regression [4] is the community-standard tool for fitting and uncertainty-quantifying fusion profile data [5, 6] — applied to KSTAR's own CES T_i profiles with SVM-based outlier rejection [7] — and monotone interpolants such as PCHIP [3, 8] are preferred over natural cubic splines [9] near transients because splines exhibit Gibbs-like ringing at ELM- and sawtooth-like steps [10]. Irregularly sampled series admit exact AR-type null models (IAR/CIAR) [11, 12]. We use this ladder—persistence, linear interpolation [13], PCHIP, local AR—as pre-registered baselines, with PCHIP as the headline comparator, and Murphy's MSE skill score [14] as the metric. Our contribution is orthogonal to this literature: rather than a better CES-only interpolant, we test whether *other diagnostics* add information beyond any CES-only method, by benchmarking a causal multimodal model against future-using interpolation.

Neural diagnostic inference in fusion. Neural diagnostic inference has three main lines, and it is instructive to ask of each: *what happens when the target measurement is absent?* The oldest line accelerates spectral fitting—NN analysis of JET CX spectra [15, 16] and fast T_i estimation from EAST CXRS spectra [17]. These gain orders of magnitude in speed over least-squares fits, but by construction they require measured spectra at the timestep: a missing CES exposure leaves them with no input at all. A second line replaces one inference pipeline with another diagnostic's: Bayesian-model emulation at W7-X [18], profile shapes from scalar parameters on NSTX-U [19], and—closest to our targets—T_i and rotation profiles from an x-ray crystal spectrometer on EAST [20]. Lin et al. predict exactly our two quantities, but from another Doppler spectrometer, i.e. the same physical information channel with the same photon-statistics limits; the mapping is instantaneous and inherits the source diagnostic's availability. A third line builds end-to-end surrogates for slow reconstruction codes: bolometer tomography [21], line-integral diagnostics [22], safety-factor profiles [23], real-time equilibria [24], and—on KSTAR itself—NN Grad–Shafranov solvers [25, 26] and rapid EPED-parameterized profile reconstruction [27] — the latter consuming CES as an *input* and explicitly naming rotation as future work. These address analysis *latency*, not measurement *availability*: they accelerate inversions of data that exist. Two works sit closer to the availability question. Actuator-conditioned profile forecasting on DIII-D [28] is causal and history-driven like our model, but forecasts from *dense* measured profiles for control, rather than recovering a sparse channel. RTCAKENN [29] reconstructs kinetic profiles (including impurity temperature and rotation) in real time and keeps operating when Thomson/CER inputs drop out, via dropout training; missingness there is a robustness condition handled by zeroing inputs, whereas in our setting the target's own sparsity is the problem itself, tracked explicitly through per-target observation flags and irregular-time features. Across all three lines the common assumption is that the information source is available when the model runs; recovering a diagnostic at timesteps where *its own* measurement is missing, from other channels plus its past, falls outside their scope.

Cross-diagnostic reconstruction and temporal densification (closest prior work). A recent family reconstructs one diagnostic from others. GP-based imputation restores faulty magnetic sensors on KSTAR in sub-ms [30]; ML detects and compensates corrupted sensors [31]; autoencoders fuse redundant density diagnostics for resilience [32]; time-series extrinsic regression fills missing electron-temperature data on EAST [33]; and FusionMAE performs masked-channel "virtual backup diagnosis" across 88 channels on HL-3 [34]. Closest to our framing, Diag2Diag synthesizes Thomson-scattering T_e/n_e at fluctuation-diagnostic rates from other simultaneous diagnostics on DIII-D [35], and temporal super-resolution of Thomson profiles from fast radiation diagnostics has been demonstrated on COMPASS [36]; PanoMHD models multimodal signal dynamics with a causal transformer for control [37]. Our work differs from this family on four axes, each verifiable from the cited works' stated scope: (i) *target channel*—every member reconstructs electron quantities or generic dense channels (Thomson T_e/n_e in 35 and 36; ECE-type T_e in 33; arbitrary masked channels in 34), and where CER/CES appears at all it is an *input* [35], never the sparse target; (ii) *causality*—Diag2Diag and the COMPASS study are simultaneous, memory-less mappings, and FusionMAE reconstructs within a window, whereas our model conditions on the target's own irregular past and never reads the future; (iii) *evaluation bar*—none benchmarks against *future-using* interpolation of the target channel, which for gap-filling is the incumbent method to beat, under a pre-registered shot-clustered protocol; and (iv) *information analysis*—reconstructability is assumed rather than tested per target, whereas the $T_i \leftrightarrow V_{\text{rot}}$ asymmetry is our central finding. In sum, this work continues the programme this active family has established—reconstructing an absent measurement from the diagnostics that are always on—and extends it in three directions: from electron channels to a *sparse ion channel*, from simultaneous memory-less mappings to *causal estimation conditioned on the target's own irregular past*, and from assumed reconstructability to *per-target tests against pre-registered baselines*. To our knowledge the conjunction of these three—causal temporal gap-filling of sparse CXRS ion measurements—has not yet been addressed; we present it as the family's natural next step rather than as a departure from it.

Irregular-time and masked multitask learning. Our design borrows standard ingredients from the irregular time-series literature—observation masks and decay-aware recurrence (GRU-D [38]), attention over continuous-time embeddings (mTAN [39]), and latent-ODE dynamics [40]—together with multitask learning under partial labels [41], late-fusion multimodal architectures [42, 43], pre-LayerNorm transformers [44], and attention pooling [45]. The "nowcasting" framing follows its use in machine-learning weather prediction [46, 47]. The novelty is not any single ingredient but the leakage-hardened combination under a physics-adverse evaluation protocol.

3 Data and problem setup

3.1 Dataset

We use 641 KSTAR discharge (shot) files, shots 30801–32751, selected by the data provider for hardware consistency and for H-mode ELM-suppression (RMP) phases, and aligned to a common 10 ms grid: 247,207 rows in total (median 339 per file). Each row provides BES [48] (9 channels), ECEI [49] (4 channels), Mirnov coils (2 channels), time, and the two CES targets T_i and V_{rot} [1]. On this grid T_i is missing in 8.2% of rows and V_{rot} in 23.9%, independently (Fig. 1). Rows are grouped into contiguous segments at time gaps ≥ 0.5 s

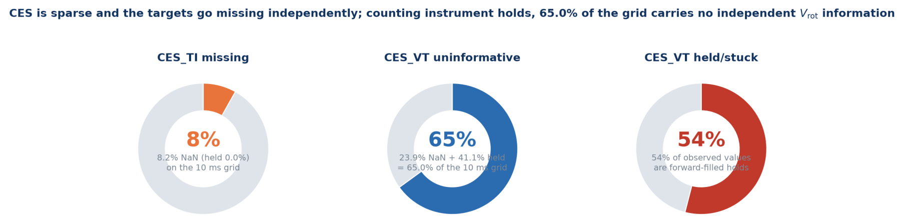

Figure 1: CES missingness structure on the 10 ms grid. T_i and V_{rot} go missing independently (8.2% / 23.9%), motivating per-target masking rather than row filtering; counting instrument holds, 65.0% of the grid carries no independent V_{rot} information.

(a threshold that sits in the valley of a bimodal delta distribution: only 82 of the ∼247k inter-row deltas fall in $(0.1, 0.5)$ s; within a segment 99.4% of steps are 10 ms): a typical file holds **one main segment** (median 301 rows ≈ 3.0 s; 10th–90th percentile 1.3–7.0 s; 28 files have two and 20 have none longer than 10 rows) plus a few isolated single rows (1,279 across the set, typically the first row at $t = 0$) separated from it by gaps of median 6.3 s. No arm—neither an interpolant nor a model's input—bridges a gap.

3.2 Two framings of a training example

The *windowed* framing (our control, §4) builds a sample at target index t from the W preceding grid rows: `bes` $(W, 9)$, `ecei` $(W, 4)$, `mc` $(W, 2)$, `time_features` $(W, 4)$, `ces_history` $(W, 4)$, the normalized target $[T_i, V_{\text{rot}}] \in \mathbb{R}^2$, and a per-target observation mask $m \in \{0, 1\}^2$. The four time features—lookback seconds, inter-row delta seconds, and their $\log(1+x)$ transforms—expose the irregular observation pattern explicitly (the reliability of a past CES value depends strongly on whether it is 10 ms or 200 ms old); the four CES-history channels are the previous normalized T_i, previous normalized V_{rot}, and one observation flag per target. W=2 throughout (§6.8).

The *full-grid sequence* framing (our main model) keeps every input-complete row of a contiguous segment—labelled or not—as context, because the fast diagnostics have no missing values. Each step carries 22 strictly causal channels: the 15 z-scored fast channels, $\log(1+\Delta t_{\text{prev}})$, and per target the last observed value carried forward from rows $< t$, $\log(1+\text{staleness})$, and a has-any-past-observation flag. Targets are z-scored with a parallel observed mask, and the sparsity is handled entirely in the loss. Nothing in either framing lets a model read the target row's own value.

3.3 Design principles

No fabricated labels. We never impute targets to create training rows. In the windowed framing a row is kept iff its diagnostic inputs are complete and *at least one* CES target is genuinely observed; in the sequence framing unlabelled rows contribute context only.

Per-target masked loss. The loss is masked per target, $\mathcal{L} = \sum_{k \in \{T_i, V_{\text{rot}}\}} m_k\,(\hat{y}_k - y_k)^2 / \sum_k m_k$, so a row with only one observed target still supervises that target. A both-targets-required filter would silently discard ≈28% of labelled rows; removing it is a pure data gain.

Leakage hardening. (i) *File-level splits*: adjacent rows within a shot are strongly autocorrelated, so splits are made at the shot-file level and pinned to disk; (ii) *train-file-only normalization*: all z-score statistics (NaN-aware for the sparse targets) are estimated on training files only—the sequence model additionally standardizes each discharge's fast diagnostics by that discharge's own mean and variance, which uses no other discharge and no target (§6.6 shows why this matters); (iii) *target-step masking*: the target row's own values and observation flags never enter the input.

3.4 Data-quality audit 1: held (forward-filled) V_{rot} values

54% of observed V_{rot} values are instrument holds: bit-identical repeats of the previous observation within a contiguous time block (runs up to 1,214 rows; 499/641 files affected). These arise because the native V_{rot} cadence is slower than the row cadence; they are not independent measurements. V_{rot} is recorded to five decimal places with a minimum spacing of 4×10^{-5} between distinct values, so "consecutive-equal" has no measurable false-positive channel; T_i is unaffected (1 held row in 226,991). Under the confirmed protocol holds are removed *everywhere*—from the supervision targets, from the history/carried inputs and their observation flags, from the normalization statistics, and from the interpolation anchors of every baseline—so no arm can score a forward-fill for free. Every number in this paper is on genuine measurements only; the ledger of the project records the earlier hold-inclusive convention and the paired retrain that showed holds also contaminate *training* (a forward-filled staircase teaches a model that copying history is near-optimal), which is why hold-free is the protocol rather than a sensitivity row.

3.5 Data-quality audit 2: T_i spectral-fit failures and the two populations

Observed T_i has p99 = 2,089 eV, p99.9 = 9,601 eV and a maximum of 14,984 eV: the far tail is not plasma physics but failed spectral fits. **1,197 rows (0.53%) exceed 3 keV**, forming 951 runs in 274 discharges; 85% of the runs are single rows, only 2% last five rows or more, and the median run peak is 13× the mean of its observed neighbours (IQR 6–26×). Such rows are unpredictable for every method, and—worse for a fair comparison—a spike *used as an interpolation anchor* poisons the interpolant at its neighbours. Two responses are defensible and each is open to a criticism: treating >3 keV as missing in every arm removes rows no method can predict but invites "the hard rows were removed", while keeping them handicaps the offline baseline by feeding it spiked anchors. We therefore pre-registered **two co-primary populations**: *cut* (values >3 keV treated as missing at load time, identically in supervision, history inputs, normalization statistics and every baseline's anchors) and *inclusive* (no cut). **A claim is stated without qualification only if it holds in both**; a result that holds in one is reported with its population named. The threshold is immaterial across 2.5–4 keV (§6.11); the value cut is a physically motivated proxy (">3 keV is not a KSTAR ion temperature") pending CES fit-quality metadata, and it is one-sided—the same audit finds single-sample *downward* dips (4,965 rows $\leq \frac{1}{2}$ both neighbours) that no value cut touches, and the cut removes only 19% of the $\geq 2\times$ upward outliers. V_{rot} has fit-failure spikes too (119 rows above 1,000 km/s in 16 shots, 101 of them one block in a single discharge); the protocol leaves V_{rot} uncut and every persistence-anchored V_{rot} comparison reports the share of squared error those rows carry.

4 Model

The main model, `seq_v2`, is a **357,570-parameter** causal sequence nowcaster; the windowed model it replaces is a 201,258-parameter late-fusion network and is retained as the paired control. The bottleneck in this problem is information, not capacity (§6.9), and both models were reached under a test-frozen selection protocol (§7).

Full-grid causal sequence backbone. `seq_v2` runs two independent causal LSTMs over the 22-channel grid sequence of a segment (§3.2). The T_i branch (2 layers, 160 units) reads the full state—fast diagnostics, both targets' carried values, staleness flags and Δt—and the V_{rot} branch (1 layer, 64 units) reads *only the seven non-fast channels*: Δt and the per-target carried value, staleness and observation flag. Each branch ends in LayerNorm and a small GELU head; outputs are normalized $[T_i, V_{\mathrm{rot}}]$ per row and the loss is the per-target masked MSE over every labelled row of the segment. Two consequences follow from the framing. First, the model's *reach* into the past is the whole segment rather than a fixed window—exactly the quantity that separates it from every windowed variant in §6.3. Second, the routing holds at the *encoder*: a shared recurrent state would carry fast-diagnostic information into the V_{rot} head no matter how the head is wired, whereas here perturbing all 15 fast channels leaves the V_{rot} output bit-identical (§6.7). Training uses AdamW (10^{-3}), batch 16 segments, early stopping on validation masked MSE (patience 6, cap 30 epochs; the confirmatory runs stopped at 14–25 epochs).

The windowed control: observation-masked attention pooling. The control encodes each modality (BES, ECEI, MC) with its own time-aware 1-D CNN over the W=2 window, reads the CES history with a bidirectional GRU (hidden 64) whose per-timestep outputs are pooled by two independent multi-head additive attention readouts, one per target, *hard-masked* so that attention mass can land only on rows where *that* target was genuinely measured—interpolation's own inductive bias (use only observed samples) written into attention at zero parameter cost. Its T_i head fuses fast diagnostics, time and history; its V_{rot} head fuses history and time only. Everything the two models share—the data contract, hold-free treatment, per-target masking, the split, the scored population—is pinned identically, so their comparison is row-for-row paired.

The interpretable rung. `b3k8` (§6.9) keeps `seq_v2`'s framing and routing but replaces each head by an *exact* decomposition, $\hat{y} = \text{carried value} + \sum_k w_k z_k + b$, where $z \in [-1, 1]^K$ is a tanh latent from a small causal GRU (64/32 units, K=8 for T_i, 4 for V_{rot}) and the readout is zero-initialized so training starts exactly at persistence. It has 21,498 parameters, 6% of the backbone.

What the search taught. Across two rounds of controlled architecture iterations on the windowed family (selection on clean validation skill only), attention pooling was the one mechanism that survived, and capacity scaling, skip topologies and extra extractors never helped. Under the confirmed protocol the same lesson recurs twice more: observation-masked causal attention added to `seq_v2` is a small, consistently positive but non-significant gain (§7), and a five-point width sweep of `seq_v2` is flat (§6.9). That is why the model section is short and the evaluation section is not.

5 Evaluation methodology

Metric. All errors are denormalized to physical CES units and computed per target. The primary score is Murphy's skill [14] against the pre-registered headline baseline, $\text{skill}_{\text{PCHIP}} = 1 - \text{MSE}_{\text{model}}/\text{MSE}_{\text{PCHIP}}$ (positive = model better; 0 = tie). Every arm (model and all baselines) is scored on the *identical* (file, row) sample set under the same per-target keep mask, verified bit-identically on the population keys before any paired comparison. Interpolants exclude the target's own value (no leakage). Neither the interpolants nor a model's input bridges a segment boundary; where a neighbour beyond the boundary would be needed the interpolant predicts the persistence value instead (no population shrinkage; PR2).

Baseline ladder. Persistence and a local past-only AR are the causal references of the original pre-registration; linear, PCHIP and a Gaussian process (exact Matérn-3/2 + white noise, local nearest-16+16 fit, per-sample marginal-likelihood grid selection, inheriting the PR2 fallback) read the future. To the causal side we add a **causal GP**: the same GP restricted to the 16 past neighbours, with the same NaN conditions so the scored population does not move. It is decisively the strongest deployable baseline (seed-42 T_i RMSE 164.3 against persistence's 197.2, cut population), and "beats every deployable causal method" is judged against it, not against persistence.

Three-way split; selection never sees TEST. Files are split train/val/test; the test files of every split were reserved *before* any architecture search and are never read during model selection. The canonical (seed-42) test set contains n = 32,589 observed T_i rows across 96 shots and n = 10,463 genuine V_{rot} rows across 60 shots in the cut population (32,721 / 10,461 inclusive); the four splits (seeds 42, 1, 7, 123) hold 32.6–35.9k T_i rows over 96 shots and 10.5–14.5k V_{rot} rows over 60–66 shots each. Every number in this paper is read from frozen run directories by a single collection script, so the text, the tables and the figures cannot drift apart.

Pre-registration. Rules fixed in writing before the corresponding test numbers were viewed: **PR1** the headline comparator is PCHIP (chosen a priori for monotonicity/no ringing at transients; the full ladder is also reported); **PR2** interpolation predicts everywhere the model is scored, with persistence fallback where no future neighbour exists—and the fallback rate is reported (0.3–0.4% of scored T_i rows, but **40–44% of scored V_{rot} rows**, so "vs. PCHIP" for V_{rot} is two-fifths "vs. persistence"); **PR3** a minimum test-set floor (≥15 shots, ≥3,000 observed T_i samples; met everywhere); **PR4** significance means a shot-clustered bootstrap 95%

Table 1: Physical-unit RMSE ladder, canonical held-out test split (seed 42), cut population, genuine measurements only (n = 32,589 T_i, 10,463 V_{rot}). Lower is better. The three light rows read the future. Inclusive population: `seq_v2` 363.0 / 23.7, PCHIP 412.4 / 30.2, causal GP 394.6 / 28.8, persistence 478.0 / 33.4 (T_i / V_{rot}); the fit-failure spikes more than double every T_i RMSE without changing the order.

Arm	Information access	T_i RMSE (eV)	V_{rot} RMSE (km/s)
`seq_v2` (nowcaster)	fast diag. + past CES, whole segment	**157.8**	**23.6**
Windowed control (W=2)	fast diag. + 2 rows of past CES	169.2	26.1
Causal GP	past 16 CES neighbours	164.3	28.8
Persistence	last observed CES	197.2	33.4
AR (local, causal)	past CES only	472.2	51.0
GP (offline)	past + future CES	153.8	24.7
Linear interpolation	past + future CES	169.8	29.0
PCHIP interpolation	past + future CES	173.6	30.2

CI on the skill excluding zero. To these the confirmed protocol adds: hold-free training and scoring; W=2 by the plateau-minimal rule of a 24-run sweep (§6.8); a per-file sample cap of 500 in the windowed family; the two co-primary populations of §3.5; and *model-decision rules committed before TEST scoring* for the backbone gate, the one attention candidate, the interpretable rung and the width sweep (§7). Threshold sensitivity (2.5/3/4 keV) is reported.

Shot-clustered paired bootstrap. Adjacent CES rows within a discharge are strongly correlated; treating samples as independent would grossly understate uncertainty. Per-sample paired errors $\mathrm{SE}_{\mathrm{model}} - \mathrm{SE}_{\mathrm{PCHIP}}$ are aggregated by shot, and whole shots are resampled with replacement (10,000 resamples, fixed seed). The resulting CI reflects genuine shot-to-shot generalization; its cost is that the effective sample size is the number of shots (≈96 for T_i, 60–66 for V_{rot}), which bounds statistical power throughout. Model-vs-model comparisons (backbone vs. control, rung vs. backbone) use the same paired bootstrap on the same rows.

The MNAR problem, and what we do about it. Truly missing points have no ground truth, so skill is measured by masking *observed* points. Because CES missingness is MNAR (drop-out at low S/N, ELMs, transitions), observed-point skill is an optimistic estimate of performance at genuinely missing points. §6.5 reweights the scored points onto the covariate distribution of the *actually missing* ones and reports how much of the skill survives, together with what fraction of the missing population is in the analysis' domain at all.

6 Results

6.1 The nowcaster dominates every causal baseline

Table 1 shows the physical-unit RMSE ladder on the canonical test split. `seq_v2` has the lowest RMSE for both targets among all methods that do not read the future, beats the causal GP—the strongest deployable arm—by 4% (T_i) and 18% (V_{rot}) in RMSE, and edges the future-using interpolants; the offline GP, which averages noisy anchors rather than passing through them, is the one arm it ties (T_i 153.8 vs. 157.8). This is the most robust result in the paper: for any online setting, where future CES is by definition unavailable, the sequence nowcaster is the clear winner.

6.2 Headline: T_i beats future-using interpolation on four independent splits, in both populations

Table 2 reports the backbone on the four independent held-out splits (seeds 42, 1, 7, 123—the latter three never used in any selection step) in the two co-primary populations, against PCHIP and against the causal GP; Fig. 3 shows the PCHIP intervals.

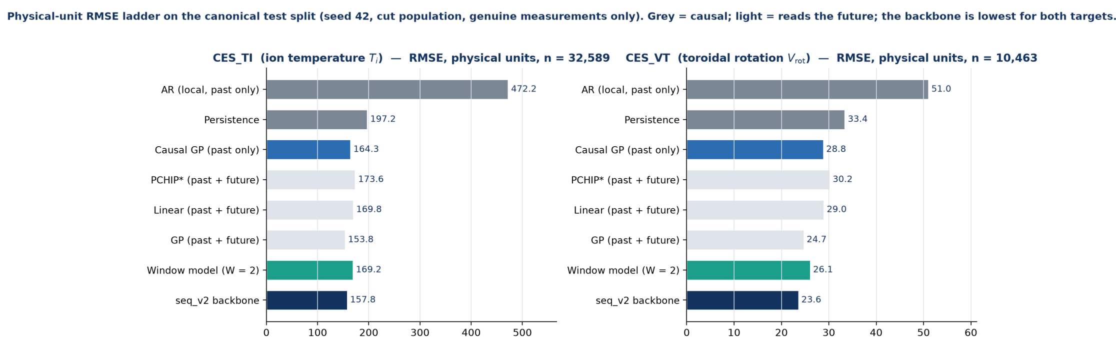

Figure 2: RMSE ladder for the seed-42 test split (cut population, genuine measurements): the causal nowcaster attains the lowest RMSE among deployable methods for both targets and edges the future-using interpolants. Same numbers as Table 1; both are generated from the same frozen artifacts.

Table 2: Held-out test skill$_{\mathrm{PCHIP}}$ of `seq_v2` with shot-clustered 95% CIs (10,000 resamples), both populations; the last column is the paired skill against the causal GP on the same rows. Bold = PASS (CI excludes 0).

Target	split	cut: skill$_{\mathrm{PCHIP}}$ (95% CI)	inclusive: skill$_{\mathrm{PCHIP}}$ (95% CI)	vs. causal GP (cut / incl.)
T_i	42	+**0.174** [+0.097, +0.236]	+**0.225** [+0.109, +0.293]	+**0.078** / +**0.154**
	1	+**0.248** [+0.188, +0.295]	+**0.238** [+0.153, +0.302]	+**0.133** / +**0.169**
	7	+**0.257** [+0.199, +0.302]	+**0.292** [+0.232, +0.344]	+**0.138** / +**0.123**
	123	+**0.264** [+0.188, +0.320]	+**0.316** [+0.186, +0.392]	+**0.105** / +**0.149**
V_{rot}	42	+**0.390** [+0.077, +0.591]	+**0.384** [+0.066, +0.586]	+**0.331** / +**0.324**
	1	+0.183 [−0.028, +0.280]	+**0.195** [+0.000, +0.286]	+0.130 / +**0.143**
	7	+0.135 [−0.358, +0.269]	+0.132 [−0.362, +0.266]	+0.020 / +0.018
	123	+0.305 [−0.049, +0.437]	+0.304 [−0.065, +0.433]	+**0.134** / +0.132

- **T_i: PASS on all four splits, in both populations.** skill$_{\mathrm{PCHIP}}$ = +0.17 to +0.26 (cut) and +0.23 to +0.32 (inclusive); every shot-clustered CI excludes zero, and every one of the eight population–split cells also beats the *causal GP* (+0.08 to +0.17) and persistence (+0.36 to +0.46). This is the unqualified claim of the paper. Against the offline GP the backbone *ties* (point estimates −0.05 to +0.11, significant on 1/8 cells), so the observed-population claim reads "beats the pre-registered interpolants and every deployable causal method, and matches a future-using, per-sample-tuned smoother".
- **V_{rot}: a tie, not a win.** Point estimates are positive on every split in both populations, but PR4 passes on 1/4 (cut) and 2/4 (inclusive)—seed 42 in both, seed 1 at [+0.000, +0.286] inclusive. Against persistence V_{rot} passes on 3/4 splits in both populations (+0.30 to +0.50; seed 7's CI touches zero) and against the causal GP on 2/4. One or two significant splits out of four is what noise would produce, so we claim no rotation win.
- **Why the inclusive numbers are higher.** Every arm scores *better* against PCHIP without the cut (backbone mean +0.268 vs. +0.236; control +0.238 vs. +0.173), because the spikes poison the interpolation anchors: a learned model discounts a spiked carried value where an interpolant cannot. §6.7 quantifies this component; it is the reason neither population is quoted alone.

6.3 The full-grid framing against the windowed control: the backbone gate

Was the sequence framing worth adopting? The decision was pre-registered as a gate with four conditions and scored once (cut population, the population under which internal model selection is done): a 4×4 grid of split seed × initialisation seed, each run paired row-for-row against its split's W=2 windowed control. Paired T_i skill was **positive on 16/16 runs and individually significant on 13/16**; the per-split initialisation means are +0.129, +0.059, +0.078, +0.058 (initialisation spread ≪ split spread); the pooled mean is +**0.081**

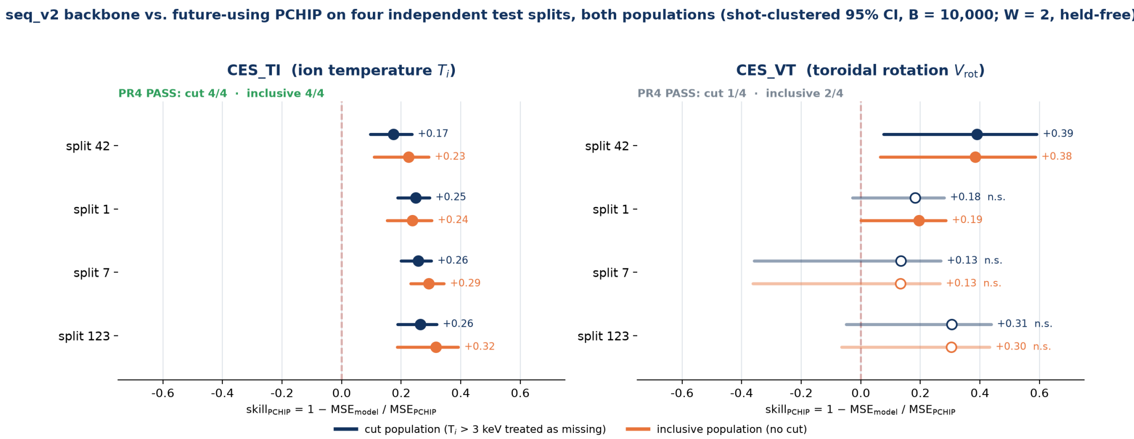

Figure 3: Forest plot of `seq_v2`'s $\text{skill}_{\text{PCHIP}}$ with shot-clustered 95% CIs across four independent held-out splits, both populations. T_i excludes zero on all eight cells; V_{rot} on three.

Table 3: Gap-stratified skill, four test splits pooled, shot-clustered 95% CIs, `seq_v2`. Bold = CI excludes zero. Rows with fewer than 50 samples are not scored.

Δt	n (cut / incl.)	shots	cut: vs. PCHIP	inclusive: vs. PCHIP
T_i				
≤ 15 ms	134,546 / 135,317	301	**+0.239** [+0.197, +0.274]	**+0.299** [+0.244, +0.347]
> 15 ms	3,422 / 3,334	265 / 263	**+0.268** [+0.187, +0.337]	**+0.206** [+0.108, +0.290]
> 45 ms	460 / 429	104 / 101	**+0.267** [+0.092, +0.414]	−0.004 [−0.304, +0.246]
V_{rot}				
≤ 15 ms	51,689	197	**+0.233** [+0.020, +0.318]	**+0.233** [+0.020, +0.317]
> 15 ms	466 / 456	130	**+0.418** [+0.104, +0.680]	**+0.432** [+0.128, +0.696]
> 45 ms	14	7	—	—

with run-cluster 95% CI [+0.067, +0.096]; a budget-equalized arm (fixed 10 epochs, final weights, no validation selection) keeps the sign on 4/4 splits (+0.063, +0.033, +0.045, +0.030), so the gain is architectural rather than a training-budget effect; and no run showed a significant V_{rot} deficit (8/16 significant wins). All four conditions held. On the four confirmatory splits reported everywhere else in this paper the same comparison reads +0.130, +0.058, +0.062, +0.044 (cut) and +0.053, +0.024, +0.047, +0.029 (inclusive), positive on 8/8, significant on 2/4 in each population.

The gate also settles *what* the framing buys. The windowed control ties the causal GP (significant on 1/4 splits, cut) and the sequence backbone beats it on 4/4 in both populations: the reach into the whole past of a segment is what clears the strongest deployable baseline. Its cost is negative—the backbone trains in a fraction of the windowed family's time because there is no per-sample window assembly or combinatorial augmentation.

6.4 Where the skill lives: gap regime

Stratifying by Δt, the time since the last observed CES value, ≥96% of scored targets lie in the adjacent regime ($\Delta t \leq 15$ ms); per split the wide bins hold only hundreds of samples, so we *pool the four test splits* and cluster the bootstrap on the physical discharge (Table 3). As Δt grows PCHIP's task gets *easier*—it interpolates between two real observations that straddle the gap—while ours gets harder, so this is the regime where a causal method is least expected to win.

The T_i win is not confined to adjacent history. Beyond 15 ms the backbone beats future-using PCHIP in both populations (+0.268 and +0.206, both CIs above zero) and beats persistence there by +0.40 and

Table 4: T_i skill of `seq_v2` reweighted from the observed population onto the genuinely-missing, in-domain one (shot-clustered 95% CIs at fixed stratum weights). Bold = CI excludes zero; *unweighted* is the headline of Table 2.

Population	split	vs. PCHIP		vs. persistence	
		unweighted	missing-matched (95% CI)	unweighted	missing-matched (95% CI)
cut	42	+0.174	+0.140 [−0.071, +0.290]	+0.360	**+0.398** [+0.278, +0.518]
	1	+0.248	**+0.164** [+0.062, +0.250]	+0.406	**+0.366** [+0.299, +0.448]
	7	+0.257	+0.203 [−0.024, +0.346]	+0.403	**+0.310** [+0.227, +0.398]
	123	+0.264	**+0.283** [+0.069, +0.392]	+0.415	**+0.381** [+0.246, +0.464]
inclusive	42	+0.225	**+0.140** [+0.033, +0.250]	+0.423	**+0.443** [+0.141, +0.623]
	1	+0.238	**+0.217** [+0.086, +0.319]	+0.436	**+0.383** [+0.273, +0.536]
	7	+0.292	**+0.167** [+0.067, +0.265]	+0.406	**+0.283** [+0.172, +0.380]
	123	+0.316	**+0.221** [+0.050, +0.320]	+0.459	**+0.337** [+0.164, +0.455]

+0.43. Beyond 45 ms it still wins under the cut (+0.267) and *ties* in the inclusive population (−0.004, CI [−0.30, +0.25]): 429 rows over 101 shots is exactly where a handful of spike-anchored rows can dominate a stratum, and we report the tie as such.

V_{rot} beats interpolation in the one place it is hardest for interpolation. The global V_{rot} tie is a tie in the adjacent regime; over $\Delta t > 15$ ms the backbone beats PCHIP in both populations (+0.418 and +0.432, 130 shots)—the single positive unconditional V_{rot} verdict in the paper—and beats persistence at ≤ 15 ms (+0.39). The widest bin has 14 rows and is not scored.

6.5 How much of this survives at the points that are actually missing?

Everything above is measured on CES points that happen to be *observed*. We can never measure error where there is no ground truth—but we can measure how skill varies over covariates that are computable at observed *and* missing rows, and reweight. Two covariates carry the mechanism: Δt since the last genuine observation, binned at 15/25/45 ms, and the input-only local-activity flag of §6.10, which is defined at a missing row because it is computed from a neighbour set that excludes the row itself. We post-stratify on $\Delta t\times$ activity, drop any stratum with fewer than 30 scored samples, and report the surviving coverage next to every estimate; the weight grid is population-consistent (the cut is applied to it too). The assumption—that within a stratum missing and observed rows are exchangeable—is stated, not assumed away.

First, a scope result. The analysis' domain is "a genuine observation of that target within two rows", the W=2 criterion. Of the genuinely missing T_i rows, **54–68% are in-domain** per split; of the genuinely missing V_{rot} rows, **only 4–6%** are, because V_{rot} gaps are long blocks (coverage of surviving strata 0.99 for T_i, 0.73–0.76 for V_{rot}). The reweighted V_{rot} row therefore answers a question about one-twentieth of the missing mass and we draw nothing from it.

Second, the two comparisons come apart—and the split is the useful thing. Against **persistence—the baseline an online system actually runs—the T_i advantage survives reweighting on all four splits in both populations** (+0.28 to +0.44, every CI excluding zero); the MNAR correction costs at most 0.12 of skill and on some splits nothing. Against **future-using PCHIP** the reweighted point estimates stay at +0.14 to +0.28 but the wide fixed-weight CIs pass through zero on two cut-population splits (2/4 significant; 4/4 inclusive). Missing points sit at larger Δt, which is where a two-sided anchor helps most and where the reweighting bootstrap is thinnest. The claim the evidence supports at the points that matter is therefore: *at genuinely missing, in-domain timesteps the nowcaster is significantly better than any causal CES-only method in both populations, and better than offline interpolation with a population-conditional significance (2/4 cut, 4/4 inclusive).* The windowed control behaves the same way (2/4 and 4/4 vs. PCHIP; 4/4 vs. persistence).

Table 5: Temporal (campaign) split, four initialisations, TEST block of 97 later discharges. Entries are skill$_{\mathrm{PCHIP}}$; **bold** = PR4 PASS. The last column is the paired T_i skill of `seq_v2` against the windowed control on the same rows.

Population	arm	T_i vs. PCHIP (init 42 / 1 / 7 / 123)	PASS	T_i vs. causal GP	`seq_v2` − control
cut	windowed control (OFF)	+0.027 / **+0.091** / −0.001 / **+0.061**	2/4	0/4	—
	control, per-shot std. (ON)	**+0.103** / **+0.107** / **+0.094** / **+0.107**	4/4	—	—
	`seq_v2`	**+0.187** / **+0.174** / **+0.181** / **+0.177**	4/4	4/4 (+0.11 to +0.12)	**+0.164** / **+0.091** / **+0.182** / **+0.124**
inclusive	windowed control (OFF)	+0.014 / +0.047 / +0.055 / +0.089	0/4	0/4	—
	`seq_v2`	**+0.173** / **+0.202** / **+0.198** / **+0.184**	4/4	4/4 (+0.13 to +0.16)	**+0.161** / **+0.163** / **+0.151** / **+0.104**

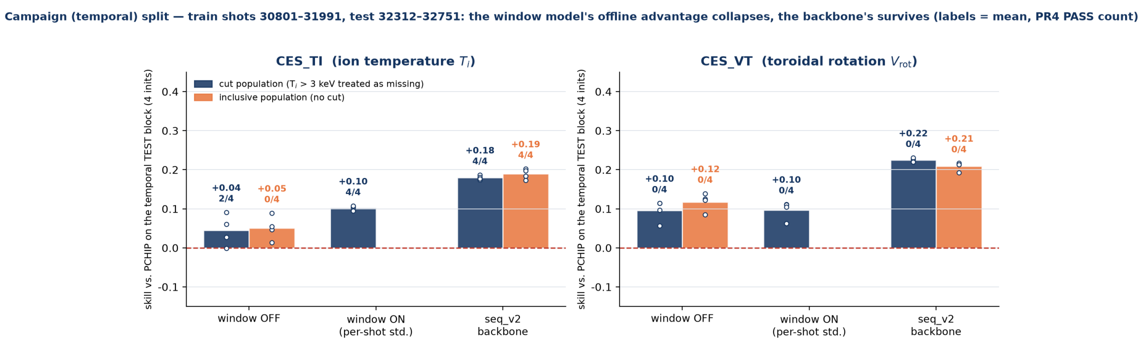

Figure 4: Campaign (temporal) split: the windowed control's advantage over interpolation collapses, per-shot standardization repairs it, and the sequence nowcaster keeps it in both populations. Bars are 4-initialisation means with per-initialisation points and PR4 counts.

6.6 Does it survive a campaign shift?

The second thing deployment does that a random split does not is move forward in time. We order all 641 discharges by shot number (30801–32751) and cut strictly in time: **train 416 shots [30801, 31991], val 128 [32002, 32310], test 97 [32312, 32751]**, no test shot preceding any train shot. Everything else is pinned to the headline protocol, so the only changed variable is the split rule; the split is fixed, so the four runs vary the *initialisation* seed only, and we say so rather than presenting them as four splits. Three arms per population: the windowed control as trained for the headline (train-file-only normalization), the same control with per-shot standardization of the fast diagnostics (cut population), and `seq_v2` (Table 5, Fig. 4).

Three things in that table are worth stating explicitly. First, the *windowed* model's advantage over offline interpolation does not survive the shift: 2/4 (cut) and 0/4 (inclusive) initialisations, and 0/4 against the causal GP. Second, its named repair works: standardizing each discharge's fast diagnostics by *its own* mean and variance at load time (targets untouched) takes the cut-population gate from 2/4 to 4/4, paired +0.078, +0.018, +0.095, +0.049 (3/4 significant), and leaves V_{rot} unchanged (−0.003 to +0.008)—the expected negative control, since the V_{rot} head never sees the fast diagnostics. Third, and this is the result of the section: **the sequence nowcaster does not collapse**. It beats PCHIP on 4/4 initialisations in both populations (+0.17 to +0.20), beats the causal GP on 4/4 in both (+0.11 to +0.16), and beats the windowed control on 8/8; its V_{rot} beats persistence on 4/4 in both populations (+0.31 to +0.34) where the windowed control's does on 0/4, and its V_{rot} margin over the control is significant on 8/8 (+0.06 to +0.18).

Why the control degrades—measured, not asserted. What fails to transfer should be the *fast-diagnostic* pathway, since that is what buys the edge over CES-only interpolation, while the CES-history pathway sees the same physical quantity in the same units in both periods. Measuring the train→test drift per channel, in the units the control was normalized in, confirms it: the median location shift is **1.22 σ for BES** and **0.53 σ for ECEI**, with scale ratios of 0.75 and 0.62—against **0.115 σ and a scale ratio of** 1.06 **for the CES targets themselves**. The fast diagnostics drift 5–11× more than the quantity being predicted. Train-file-only normalization (§3) is the correct leakage-hardening choice under a random split and is precisely what breaks under campaign shift; per-shot standardization uses only that discharge's own data, so it stays

Table 6: The two claims under stress, `seq_v2`, T_i. Entries are PR4 pass counts (cut / inclusive) with the point-estimate range.

Evaluation	vs. PCHIP (offline, future-using)	vs. causal baselines
Random split, observed points	**4/4** / **4/4**, +0.17 to +0.32	**4/4** / **4/4** vs. causal GP, +0.08 to +0.17
Reweighted to missing points (§6.5)	2/4 / **4/4**, +0.14 to +0.28	**4/4** / **4/4** vs. persistence, +0.28 to +0.44
Temporal campaign split (§6.6)	**4/4** / **4/4**, +0.17 to +0.20	**4/4** / **4/4** vs. causal GP, +0.11 to +0.16

Table 7: Evaluation-time modality ablation of the $W=2$ windowed control, TEST, both populations: $\text{skill}_{\text{PCHIP}}$ (4-split mean) and the paired change against the un-ablated run (splits 42 / 1 / 7 / 123; * = significantly worse).

Inputs	target	cut	inclusive
Full (history + fast + time)	T_i	+0.173	+0.238
History + time only (*no fast*)	T_i	−0.125; paired −0.25* / −0.38* / −0.42* / −0.43*	+0.201; paired −0.03* / −0.04 / −0.03 / −0.09*
Fast + time only (*no history*)	T_i	−2.11; paired −4.6* / −1.8* / −2.3* / −1.9*	−1.16; paired −1.5* / −3.4* / −1.2* / −1.1*
Full	V_{rot}	+0.213	+0.206
History + time only (*no fast*)	V_{rot}	+0.213; paired +0.000 ×4 (**bit-identical**)	+0.206; paired +0.000 ×4 (**bit-identical**)
Fast + time only (*no history*)	V_{rot}	−2.89; paired −5.4* / −6.3* / −1.8* / −2.3*	−3.51; paired −6.9* / −7.8* / −1.9* / −2.2*

leak-free. `seq_v2` standardizes per shot by definition—which is one of the two reasons it transfers; the other is reach (§6.3).

Two stress tests, one reading. Table 6 puts them side by side for the adopted model. The advantage over the baselines a real-time system can actually run survives both stress tests in both populations. The advantage over *future-using* interpolation survives the temporal shift in both populations and the MNAR reweighting with population-conditional significance. The version of this table for the windowed model—which the earlier draft of this work reported—had the offline advantage surviving *neither* test; the sequence framing is what changed it, and we say so rather than presenting the new column as if it had always been there.

Caveat that stays. The four campaign passes are initialisations on one temporal test block, so this is one temporal split's evidence, not four; and two of the four cut-population `seq_v2` campaign runs stopped at the 30-epoch cap.

6.7 The $T_i \leftrightarrow V_{\text{rot}}$ information asymmetry

A controlled input-modality ablation on the windowed control—one modality group zeroed *at evaluation*, no retraining, persistence and interpolants computed from the real history—explains *why* the two targets behave differently and re-grounds the routing of §4 under the confirmed protocol (Table 7, Fig. 5).

- **History is indispensable for both targets.** Without the CES history the windowed model falls to −1 to −4 against PCHIP on both targets in both populations: neither target is predictable from 100 Hz diagnostics without its own anchor.
- **T_i: the margin over interpolation is fast-diagnostic information—in the cut population.** With BES/ECEI/MC zeroed the control drops *below* the interpolator (−0.10 to −0.18; paired −0.25 to −0.43, 4/4 significant). The physical channel is collisional electron–ion coupling ($t_{ei} \propto T_e^{3/2}/n_e$): ECEI supplies T_e and BES supplies n_e structure.
- **In the inclusive population the same history-only model still beats PCHIP by +0.15 to** +0.23, and the fast channels add only 0.03–0.09 (2/4 significant): the interpolant's anchors are spiked and a learned model discounts them where an interpolant cannot. The spike-inclusive margin therefore contains a *spike-robustness* component that is not fast-diagnostic information, which is the measured reason this paper reports two populations rather than the more flattering one alone; the cut population is the one that isolates what the fast diagnostics contribute.

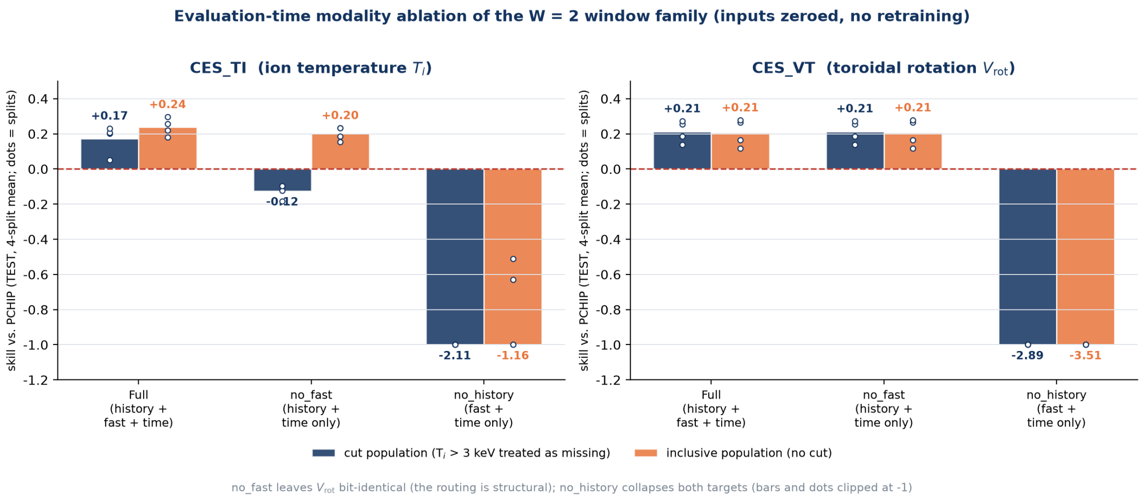

Figure 5: Evaluation-time modality ablation of the windowed control, both populations. Zeroing the fast diagnostics leaves $V_{\rm rot}$ bit-identical and removes the T_i margin under the cut; zeroing history collapses both targets.

Table 8: Window sweep, mean held-out test skill$_{\rm PCHIP}$ over 4 seeds (PASS = seeds whose shot-clustered 95% CI excludes zero). History-0 is the `no_history` ablation at W=4; W=1 is not constructible.

History obs.	W	T_i skill	PASS	$V_{\rm rot}$ skill	PASS
0	4	−0.026	0/4	−0.783	0/4
1	2	+0.238	**4/4**	**+0.206**	0/4
2	3	**+0.246**	**4/4**	+0.203	1/4
3	4 *(old default)*	+0.221	3/4	+0.190	1/4
5	6	+0.190	3/4	+0.205	1/4
7	8	+0.216	**4/4**	+0.204	2/4

- **$V_{\rm rot}$: information is entirely CES history, by construction and by measurement.** Zeroing the fast diagnostics leaves the $V_{\rm rot}$ output *bit-identical* on 8/8 population–split cells (the routing is structural), and `seq_v2`'s and `b3k8`'s $V_{\rm rot}$ branches pass the same perturbation test. Toroidal rotation is governed by momentum sources none of our inputs observe—the external NBI torque and intrinsic rotation drives [50]—and the Mirnov signals are sampled at 100 Hz, aliasing away the kHz mode-rotation frequencies that could otherwise proxy rotation (§9).

This asymmetry was predicted from physics before the ablation and is confirmed by it; the $V_{\rm rot}$ non-win is a finding about diagnostic information content, not a model failure. It also motivates the per-target routing of §4.

6.8 How much history is needed? One observation

The history window of the windowed family was originally an unexamined default (W=4). We replaced the assertion with a curve and a stated selection rule: **24 independent runs**, $W \in \{2, 3, 4, 6, 8\}\times$ seeds $\{42, 1, 7, 123\}$, plus a *history-0* point ×4 seeds, each scored on its own held-out test split with the same protocol as the headline (hold-free, per-file cap 500, no spike cut—this sweep predates the cut, and its frozen W=2 runs are the inclusive-population control of §6.2). Three controls make the points comparable: the file-level split is window-invariant (all runs evaluate the same 96 test shots per seed); the per-file cap prevents temporal-subset augmentation—which explodes from 240k samples at W=2 to 30.1M at W=8—from letting a few long-block discharges dominate; and the scored population shrinks only 1.8% from W=2 to W=8.

Table 9: Complexity ladder, held-out TEST skill$_{\text{PCHIP}}$ for T_i (4-split mean), and the paired verdicts of the pre-registered rule, both populations. The anchor+Δ model collapses onto persistence at W=2 (its slope term needs two observed history rows and never fires).

Arm	params	cut	inclusive	`b3k8` − anchor (T_i, cut / incl.)	`b3k8` − `seq_v2` (T_i, cut / incl.)
Persistence	0	−0.264	−0.288		
Anchor+Δ (named terms)	1,258	−0.261	−0.287		
`b3k8`	**21,498**	**+0.237**	+0.126	+0.35 to +0.42, 4/4* / +0.29 to +0.34, 4/4*	mean **+0.002** (all CIs $\ni$ 0) / mean **−0.194** (4/4*)
Windowed control	201,258	+0.173	+0.238		
`seq_v2` backbone	357,570	+0.236	+0.268		

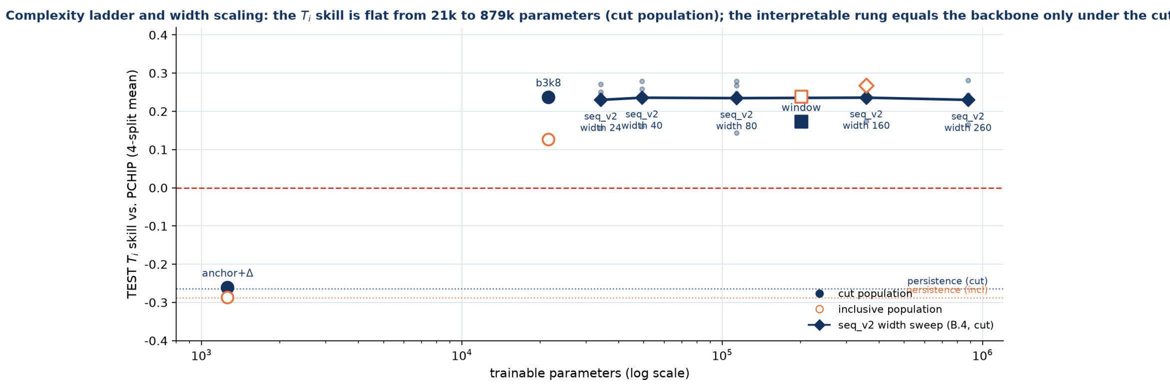

Figure 6: Complexity ladder and width sweep on one axis (TEST T_i skill vs. trainable parameters, 4-split means). Under the cut the 21k latent-bottleneck model sits on the backbone's curve and the curve is flat from 34k to 879k parameters; in the inclusive population the bounded-correction rung drops to +0.126.

History is essential, and its first observation carries essentially all of it. Removing previous-CES entirely puts T_i *below* PCHIP (−0.026, 0/4) and V_{rot} at −0.78. A *single* past observation lifts both targets to their plateau at once and the curve is flat from there: T_i spans 0.190–0.246 and V_{rot} 0.190–0.206, while the seed spread *within* a single point is 0.07–0.16, wider than the whole curve. Applying "smallest W that reaches plateau" returns $\mathbf{W = 2}$; the incumbent W=4 is below the W=2/W=3 values on both targets, so nothing in the data supports paying for a longer window on accuracy grounds. The one defensible reason to go wider is **coverage**: a windowed sample is scored only when an observed value exists inside the window, so $\Delta t > 15$ ms scored rows go 456 → 1,958 and $\Delta t > 45$ ms 14 → 135 from W=2 to W=8. That is not an argument for a longer window; it is the argument for the sequence framing, whose reach is the whole segment and whose W is not a hyper-parameter.

6.9 What does the complexity buy, and does size help? A ladder and a width sweep

A fair criticism of any learned nowcaster is that its parameters are unexplainable. The useful reply is to *measure what the complexity earns*, under the same protocol, with decision rules fixed before TEST scoring.

The interpretable rung. `b3k8` (§4) predicts $\hat{y}$ = carried value + $\sum_{k=1}^{K} w_k z_k + b$ with $z \in [-1, 1]^K$ from a small causal GRU and a zero-initialized readout, so (i) the decomposition into K+1 named terms is exact, not an attribution approximation; (ii) every number the readout can see is one of K probeable latents; (iii) the correction is bounded by $\sum_k |w_k| + |b|$ in target σ. K was the one explored variable (val only, two splits; K=8 chosen), and both arms stop by the same patience rule with a non-binding cap. Two pre-registered conditions: beat a retrained named-terms anchor+Δ model (1,258 parameters) on 4/4 splits with no V_{rot} deficit, and stay within −0.05 of the backbone's T_i skill on average.

- **Under the cut, all of the backbone's T_i skill survives compression to eight bounded numbers plus persistence.** `b3k8` beats the retrained anchor on 4/4 splits with no V_{rot} deficit, and its paired

T_i skill against the backbone is -0.009, -0.005, $+0.026$, -0.004 (mean $+0.002$, every CI containing zero); it passes PR4 vs. PCHIP 4/4 and beats the causal GP 4/4. Linear probes on TEST inputs (no targets touched) show the T_i latent encodes the last observed T_i (R^2 0.47–0.75) and the ECEI T_e proxy (0.31–0.48), distributed across dimensions, and barely encodes local BES activity (0.09–0.13) or staleness (0.01–0.07); the learned correction accounts for 25–39% of prediction variance. What the task needs is the full-grid causal state, and that state is small.

- **Without the cut the rung costs -0.16 to -0.21 against the backbone (4/4 significant) and sits below the windowed control on 3/4 splits**—not because eight latents are too few, but because a bounded correction cannot recover a spiked carried value: rows whose persistence error exceeds 2 keV are 0.6–1.3% of the inclusive test rows and carry 73–83% of `b3k8`'s T_i squared error (and 70–83% of every other arm's, PCHIP included). The rung condition still holds there (4/4 vs. the anchor); the backbone-tolerance condition is cut-conditional, and we state it so.

- **More capacity buys nothing.** Widening the `seq_v2` T_i encoder from 24 to 260 units (34k / 49k / 114k / 358k / 879k parameters; every other setting fixed; TEST once per point) gives mean T_i skill $+0.230$ / $+0.236$ / $+0.235$ / $+0.236$ / $+0.230$ (cut), paired against the 358k backbone within ±0.008 on average, the widest point significantly better on 1/4 splits, and no width down to 24 units significantly worse on $\geq3/4$; V_{rot} (branch fixed) does not move ($+0.250$ to $+0.254$). Together with the rung, this closes the model-size axis: the T_i information in {100 Hz BES/ECEI/MC + CES history + time} is exhausted by ∼50k parameters of causal recurrent state, and the remaining variance across splits ($+0.14$ to $+0.17$ on seed 42 vs. $+0.23$ to $+0.28$ on seed 123) is split variance, not capacity.

6.10 The edge concentrates in high-variability neighbourhoods

Using a target-independent, input-side proxy for local activity (large neighbour-bracket slope and/or high local CES-neighbour variance, computed *excluding* the target row), we split each TEST split into "peak" (high-local-activity; 4.1–4.6k T_i rows and 2.4–2.9k V_{rot} rows per split) and bulk rows and score `seq_v2` in each with the shot-clustered bootstrap (Fig. 7):

- T_i: peak-stratum skill$_{\mathrm{PCHIP}}$ = $+0.45$ to $+0.61$ (cut) and $+0.62$ to $+0.72$ (inclusive), **PASS on 8/8**, against an interpolator that has the future neighbour; the bulk carries $+0.09$ to $+0.20$ (cut, PASS 4/4) and $+0.06$ to $+0.19$ (inclusive, PASS 2/4). Interpolation is near-optimal on the smooth bulk; the model's value is in the active stretches, and that statement is unconditional.

- V_{rot}: peak-stratum skill $+0.54$ to $+0.79$ (point estimates positive on 8/8, PASS 2/4 in each population, against persistence $+0.75$ to $+0.86$ PASS 8/8), and ≈ 0 in the bulk (-0.07 to $+0.15$, PASS 0/8). The asymmetry is therefore *regional*: globally V_{rot} ties interpolation, but in high-activity neighbourhoods—where smooth past+future interpolation is worst—even the history-driven V_{rot} predictor adds value. We report it as a regional strength with a thin significance base, not a reversal of the global null.

6.11 Cut-threshold sensitivity

Retraining the backbone with the spike threshold at 2.5 and 4 keV (four splits each, scored in its own population) gives mean T_i skill $+0.230$ and $+0.232$ against $+0.236$ at 3 keV, PR4 4/4 vs. PCHIP and 4/4 vs. the causal GP at every threshold, and V_{rot} $+0.252$ / $+0.253$ / $+0.257$. The threshold is immaterial across the range a physicist would defend; what matters is the two populations, not where inside the tail the cut falls.

7 Model-selection protocol

Every model decision in this paper was taken on validation data or under a decision rule committed in writing before the corresponding TEST scoring, and TEST was scored once per decision.

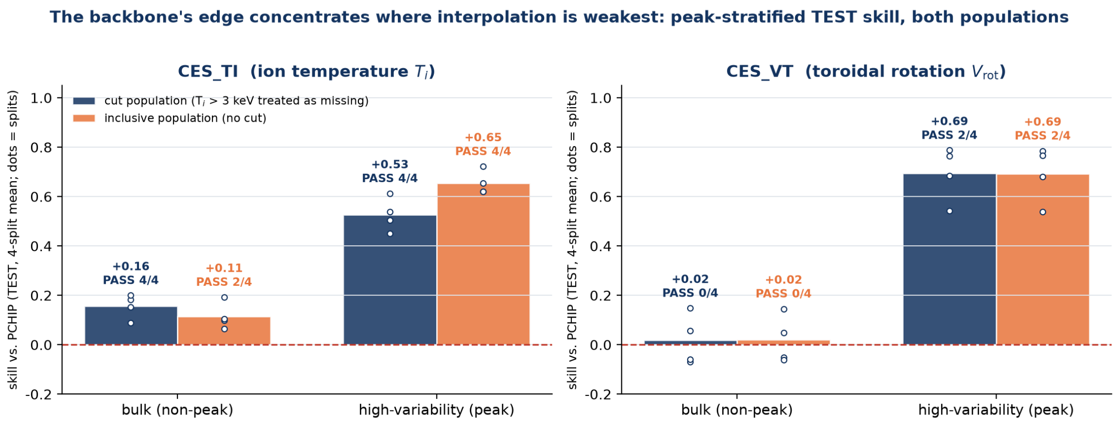

Figure 7: Peak-stratified TEST skill of `seq_v2`, both populations (4-split means, per-split points, PR4 counts). The advantage over interpolation concentrates where interpolation is weakest; V_{rot} skill lives entirely in the peak stratum.

The windowed family. Its architecture was reached by a sequence of controlled experiments, each *one* change that had to preserve the data contract, scored on the *clean, non-augmented* validation skill$_{\text{PCHIP}}$—never the augmented validation loss, which rewards smoothing exactly where interpolation is already strong—and kept iff it improved the best score so far. Its history length was then set by the sweep of §6.8, and holds were removed from training by the audit of §3.4.

The backbone gate. The sequence framing was adopted only after the four-condition gate of §6.3 (sign on 4/4 splits, pooled run-cluster CI excluding zero, sign preserved under budget equalization, no V_{rot} deficit) had been fixed and then met. Under the same rule the one architectural candidate explored afterwards—observation-masked causal attention over each target's own past observation steps, added to `seq_v2` with zero-initialized projections—was positive on 4/4 splits (+0.009, +0.013, +0.033, +0.020 vs. the backbone) but significant on 1/4 against a pre-committed bar of $\geq 3/4$, so it was *not* promoted; the backbone stayed `seq_v2`. Its validation gains had been 2/2 significant on the exploration splits, which is the usual optimism of selection-split results and the reason the promotion bar is set on TEST.

The rung and the width sweep (§6.9) had their two-branch verdicts and their descriptive readings (ceiling / knee) written down before any TEST scoring, and no backbone re-selection was allowed on the sweep by construction. The full pre-registration, its per-batch execution records and the runner scripts are in the released repository.

8 Is it deployable?

Two things stand between a skill score and a usable instrument: it has to run inside the measurement cadence, and it has to say how much to trust it.

Latency: stateful one-step inference fits on a CPU with room to spare. CES sits on a 10 ms grid, so an online gap-filler has 10 ms per step minus whatever acquisition and control already spend. A recurrent nowcaster is run online *statefully*: the hidden state is carried across the grid and each new row costs one recurrent step at batch 1, which is the number that matters. Measured as the network forward pass only (feature assembly is outside the model) over 1,000 timed calls after warm-up on an idle laptop-class machine, the `seq_v2` step costs **1.05 ms median and 1.61 ms at p99 on CPU** (16% of one grid period; p95 1.35 ms), and 1.21 / 2.31 ms on the GPU—at this size a GPU buys nothing at batch 1. The whole-segment re-run that offline scoring uses costs 2.9 ms median / 5.6 ms p99 for a 100-row (1 s) block and 6.4 / 8.9 ms for 300 rows on CPU, i.e. 35–47k rows/s of bulk reprocessing (47–63k rows/s on the GPU). The windowed control at batch 1 is slower and, on this machine, tail-heavy: W=2 3.8 ms median but 18.9 ms at p99 on CPU (7.2 / 11.6 ms on the GPU); W=4 4.0 / 8.1 ms. Two idle-machine runs of the same benchmark differed by up to 2× in absolute terms (the `seq_v2` step measured 0.51 / 0.99 ms median / p99 in the other run) because a laptop's power states move these numbers; the ordering—sequence step ≪ windowed forward pass, CPU sufficient—did not. The practical guidance is therefore: *run the stateful nowcaster on the control computer's CPU*; it leaves more than 80% of the 10 ms budget to acquisition and control.

Uncertainty: distribution-free intervals without touching the model. The community standard for fusion profile data is Gaussian-process regression, which returns a posterior; our model returns a point. Fitting a variance or quantile head would require retraining and would move every number above—so we use *split conformal prediction* instead: calibrate on the run's own validation split, change nothing about the predictor, obtain distribution-free finite-sample marginal coverage. Two variants: a single quantile (*global*) and one per Δt bin (*Mondrian*). The identical procedure is applied to PCHIP and persistence, so what is compared is interval quality, not calibration technique. At $\alpha = 0.10$, TEST, both populations:

- **The model's intervals beat both baselines' in every cell** (32/32 population × target × variant × split) by the Winkler interval score, which penalises width and misses jointly: T_i 1,272 vs. 1,554 (PCHIP) vs. 1,727 (persistence) in the cut population, 2,290 vs. 2,851 vs. 3,120 inclusive; V_{rot} 150 vs. 164 vs. 179 (global variant, 4-split means). This does not follow from better point estimates; it is a separate property.
- **In the inclusive population the model's T_i intervals are actually *wider* than PCHIP's** (half-width 224–255 vs. 211–241 eV) and still score better: the spikes inflate the miss penalty, and the model misses less. Mondrian calibration tightens every T_i arm by ≈4–5%, leaves V_{rot} unchanged, and changes no verdict.
- **Honest failure: coverage is marginal, not conditional.** T_i coverage is 0.87–0.92 against a 0.90 target (one split under-covers for every arm) and V_{rot} 0.91–0.94; per-shot coverage still ranges widely. Conformal guarantees coverage under exchangeability, and calibration and test are disjoint *discharges*; shot-level shift breaks it. For a deployed instrument this matters more than the pooled number, and fixing it needs shot-conditional calibration that the present shot count cannot support.

9 Where the remaining headroom is

The negative results in this paper are unusually specific, and specificity is what makes them useful: they do not say "this problem is hard", they say *which change would move it*. Capacity is ruled out (§6.9), longer windows are ruled out (§6.8), and the framing that buys reach has been adopted (§6.3). What remains is data, and each lever is identified by evidence already in this dataset.

1. CES fit-quality metadata would replace a proxy with a measurement. The two-population report exists because a value cut is a proxy for "the fit failed": it removes one-sample upward events that no method can predict, but it is one-sided (dips are untouched) and it catches only 19% of ≥ 2× upward outliers (§3.5). With per-sample fit χ^2 or signal level, a quality cut would replace or accompany the value cut in every arm at once, and the two populations would collapse into one. V_{rot}'s own spikes (119 rows above 1,000 km/s) would be handled by the same rule; until then any persistence-anchored V_{rot} comparison reports the share of squared error those rows carry, as §6.9 does.

2. The Mirnov information was destroyed by preprocessing, not absent from the plasma. It would be wrong to conclude from §6.7 that magnetics carry no rotation information. On the identical 10 ms grid, lag-1 autocorrelation within contiguous blocks is +0.568 for BES and +0.572 for ECEI but −0.009 for the Mirnov coils, with 82% of blocks below $|r| = 0.1$: the magnetic channel is white noise *on this grid*. That is the signature of $\mathrm{d}B/\mathrm{d}t$ oscillating at the kHz mode frequency and being decimated to 100 Hz without an anti-aliasing filter, so successive samples have random relative phase. The mode-rotation frequency—the one quantity in our diagnostic set that could proxy toroidal rotation—was therefore discarded upstream of the model, which is also why every derived MC feature we tried (integral, PCHIP integral, $|MC|$, rolling RMS) failed: information already lost cannot be recovered downstream. The fix is a preprocessing change, not a model change: per-window RMS, band power, mode number and mode-rotation frequency computed from the *raw* kHz Mirnov timeseries, routed into the $V_{\rm rot}$ branch under a pre-registered pilot-then-expand rule. This is the highest-value experiment we can name for $V_{\rm rot}$, and it is testable on archived data.

3. The actuator channel is simply absent. Toroidal rotation is set by injected torque, and no torque signal exists in this dataset. The data show the break precisely: between shots, the ECE-derived T_e surrogate correlates with T_i at $r = +0.353$ ($p = 3 \times 10^{-17}$) but with $V_{\rm rot}$ at $r = +0.024$ ($p = 0.58$), and within shots the T_i correlation keeps a consistent sign while the $V_{\rm rot}$ one is sign-random. So T_e does track the heating level, and heating level does not reach rotation—power is not torque, which depends on beam energy, tangency radius and injection geometry. Adding NBI torque (or the per-beam power and geometry from which it is computed) is a data-acquisition task, not a modelling one, and it is the only lever we can identify that addresses the *cause* of rotation rather than its autocorrelation. A positive control already exists in the literature: recurrent full-discharge simulators on DIII-D that receive the beam actuators as inputs do predict rotation's evolution [51], so the missing channel is learnable once measured.

None of these is a statement that the present result is at a ceiling. They are the places where the next measurable gain is, ordered by how cheaply they can be tried, and all three are actionable on archived KSTAR data.

10 Limitations

Statistical power. The unit of replication is the shot; with 96 (T_i) / 60–66 ($V_{\rm rot}$) test shots per split, power bounds every significance call, and per-shot squared-error differences are heavy-tailed (a few discharges dominate the bootstrap spread; in the inclusive population $\approx$1% of rows carry 70–83% of every arm's T_i squared error). **MNAR optimism.** Skill is measured on observed points only; the reweighting of §6.5 bounds the optimism for the causal comparison in both populations and for the offline comparison with population-conditional significance, and it covers 54–68% of missing T_i rows and 4–6% of missing $V_{\rm rot}$ rows. **The offline claim is bounded above by a GP tie.** The offline GP beats PCHIP and the model only ties it (1/8 cells significant), so "beats future-using interpolation" means "beats the pre-registered interpolants"; it does not extend to every offline method. It does extend to the causal GP, the strongest deployable one. **The value cut is a proxy.** Two populations bracket the fit-failure question; a fit-quality cut would settle it (§9), and $V_{\rm rot}$'s spikes remain uncut. **Campaign transfer rests on one temporal block** (§6.6): four initialisations, not four splits, and 2/4 cut-population runs stopped at the epoch cap. The repair that makes the windowed control transfer, per-shot standardization, is the offline form; a causal running (expanding-window/EWMA) estimator is the version that would ship and is not measured here. **Metric asymmetry.** Interpolants use full-shot neighbours while the windowed control sees two rows and the backbone the segment's past; this is deliberately adverse to the causal models but complicates direct interpretation. **Single device and model family.** Everything is one device, one diagnostic set and two closely related recurrent families; nothing here establishes transfer to another tokamak. **Uncertainty is calibrated marginally, not conditionally** (§8). **Latency** is a network-only measurement on one laptop-class machine, with up to 2× run-to-run variation from power states (§8); feature assembly and acquisition overhead are not included. **Scope.** The pedestal-top framing is inherited from the data selection (pedestal-top CES channels); no radial-dependence experiment is performed, and event-phase analyses (ELM cycle, L–H transitions) are future work—the high-variability analysis of §6.10 is an event-agnostic proxy.

11 Conclusion

We built a causal, leakage-hardened multimodal nowcaster for sparse KSTAR CES targets—a full-grid two-branch recurrent model whose $V_{\rm rot}$ branch never sees the fast diagnostics—and evaluated it under a pre-registered, shot-clustered, two-population protocol against a deliberately adverse bar: interpolation that sees the future, and a causal GP that is the strongest method that does not. On the observed population the model beats future-using PCHIP for ion temperature with significant, replicated skill (+0.17 to +0.32 across four independent splits and both populations), ties the strongest offline smoother, and beats every causal baseline for both targets, including the causal GP on all eight cells and including the non-adjacent regime ($\Delta t > 15\,\mathrm{ms}$).

The claim we make about deployment is now supported by both stress tests. At genuinely missing, in-domain timesteps the advantage over every causal method survives reweighting on 8/8 population–split cells; across a campaign boundary the sequence nowcaster stays ahead of PCHIP and of the causal GP on 4/4 initialisations in both populations, where the windowed control it replaced had lost its offline advantage entirely—and the difference is measured, not asserted: the fast diagnostics drift 5–11× more between campaigns than the targets, per-shot standardization repairs the control, and the sequence framing standardizes per shot and reaches the whole past by construction. An online virtual sensor competes with persistence and a causal smoother, not with an interpolant that reads the future; by that comparison the nowcaster works at the points where it matters. It emits calibrated intervals that beat both baselines' by a proper scoring rule in every cell.

We also report where it does not work and why. Rotation ties interpolation globally and its skill lives in transients; the value cut that defines one of the two populations is a one-sided proxy for fit quality; the reweighting reaches 54–68% of missing T_i rows and a twentieth of missing V_{rot} rows; the predictive intervals are calibrated marginally, not per discharge; and beyond 45 ms the inclusive population is a tie. None of these is a ceiling argument, and none is left as a shrug. Each points at a specific, testable change—CES fit-quality metadata; per-window features recomputed from the raw kHz Mirnov stream whose information 100 Hz decimation destroyed before the model ever saw it; the NBI torque channel that is absent from the dataset but is the physical cause of rotation—and the model-size axis is closed: a 21k-parameter persistence-plus-bounded-correction model matches the backbone under the cut, and a 26× width sweep is flat. Alongside these, the work contributes a corrected V_{rot} evaluation (54% instrument holds, removed everywhere), a two-population treatment of spectral-fit failures with the measurement that shows why neither population suffices alone, a full-grid causal framing whose reach is what clears the strongest deployable baseline, and a test-frozen selection protocol whose every decision rule was written down before the number it decided.

Reproducibility

Splits are pinned to disk and verified on reload; normalization statistics are train-file-only (plus per-shot input standardization for the sequence family); model selection never reads test data; bootstrap uses a fixed seed and 10,000 resamples; every quoted number is read from frozen evaluation artifacts by one collection script.

Code availability

The full pipeline, the pre-registration document with its per-batch execution records, the controlled-experiment runners behind every result reported here, the pinned train/validation/test splits, and the collector that regenerates every quoted number from the frozen evaluation artifacts are openly available at `https://github.com/iseungsang01/ml-intern-revision`. The backbone is `ces_prediction/experiments/seq/model_seq_v2.py`, the interpretable rung `model_seq_b3.py`, and the windowed control `ces_prediction/model_iter009.py` (guarded by a content hash in the test suite), so the code that produced these numbers cannot drift from the code that is released.

Data availability

The KSTAR diagnostic data underlying this study — 641 discharges, shots 30801–32751 — are not redistributed with the code. They are experimental data of the KSTAR device and are governed by the operating institution's data policy; access requests should be directed there. The repository contains everything needed to reproduce the analysis once the shot files are in place: the exact file-level split manifests, the environment configuration for every reported run, and the per-sample squared-error archives from which the bootstrap statistics are computed.

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